Codec technologies:

1. AI-Powered Chatbot

Description: Build an intelligent chatbot using Natural Language Processing (NLP) for customer support or FAQs.

Technologies: Python (NLTK, Transformers), Flask/FastAPI, SQLite

Outcome: Deploy a chatbot with contextual responses and user interaction logs.

Code:

AI CHATBOT

```
# Install required libraries before running:
!pip install flask nltk

from flask import Flask, request, jsonify, render_template_string
import sqlite3
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords

# Download NLTK data
nltk.download('punkt')
nltk.download('stopwords')

# DATABASE SETUP (SQLite)
conn = sqlite3.connect("chatbot.db", check_same_thread=False)
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS chats (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_input TEXT,
```

```
    bot_response TEXT
)
""")

def save_chat(user, bot):
    cursor.execute("INSERT INTO chats (user_input, bot_response) VALUES (?, ?)", (user,
bot))
    conn.commit()
# NLP PREPROCESSING
def preprocess(text):
    tokens = word_tokenize(text.lower())
    words = [w for w in tokens if w.isalnum()]
    return words

# CHATBOT LOGIC (FAQ BASED)
faq = {
    "hello": "Hi! How can I help you?",
    "hi": "Hello! Ask me anything.",
    "fees": "Our course fee is ₹20,000.",
    "courses": "We offer Data Analytics, AI, and ML courses.",
    "duration": "The course duration is 6 months.",
    "bye": "Goodbye! Have a great day!"
}

def get_response(user_input):
    words = preprocess(user_input)
    for word in words:
        if word in faq:
            return faq[word]
    return "Sorry, I don't understand. Please try again."
# FLASK APP
```

```
app = Flask(__name__)
# HTML FRONTEND (inside Python 😎)
HTML_PAGE = """
<!DOCTYPE html>
<html>
<head>
    <title>AI Chatbot</title>
    <style>
        body { font-family: Arial; text-align: center; }
        #chatbox { width: 50%; margin: auto; }
        input { width: 70%; padding: 10px; }
        button { padding: 10px; }
        .msg { margin: 10px; }
    </style>
</head>
<body>

<h2>🤖 AI Chatbot</h2>

<div id="chatbox"></div>

<input type="text" id="msg" placeholder="Type your message...">
<button onclick="send()">Send</button>

<script>
function send() {
    let msg = document.getElementById("msg").value;
fetch("/chat", {
        method: "POST",
        headers: {"Content-Type": "application/json"},
```

```
        body: JSON.stringify({message: msg})
    })
    .then(res => res.json())
    .then(data => {
        document.getElementById("chatbox").innerHTML +=
            "<div class='msg'><b>You:</b> " + msg + "</div>";
        document.getElementById("chatbox").innerHTML +=
            "<div class='msg'><b>Bot:</b> " + data.reply + "</div>";
    });

    document.getElementById("msg").value = "";
}
</script>

</body>
</html>
"""
# Home page
@app.route("/")
def home():
    return render_template_string(HTML_PAGE)

# Chat API
@app.route("/chat", methods=["POST"])
def chat():
    user_input = request.json["message"]
    response = get_response(user_input)

    save_chat(user_input, response)
```

```
    return jsonify({"reply": response})
if __name__ == "__main__":
    app.run(debug=True)
```

output :

```
Run started
Initializing environment
Installing packages
Running code
[nltk_data] Error loading punkt: <urlopen error unknown url type:
[nltk_data]     https>
[nltk_data] Error loading stopwords: <urlopen error unknown url type:
[nltk_data]     https>
 * Serving Flask app '__main__'

 * Debug mode: on
ModuleNotFoundError: No module named '_multiprocessing'
   line 135, in <module>
  File "/lib/python3.12/site-packages/flask/app.py", line 662, in run
    run_simple(t.cast(str, host), port, self, **options)
  File "/lib/python3.12/site-packages/werkzeug/serving.py", line 1083, in run_simple
    application = DebuggedApplication(application, evalex=use_evalex)
                  ^^^
```

```
  File "/lib/python312.zip/multiprocessing/context.py", line 72, in RLock
    from .synchronize import RLock
  File "/lib/python312.zip/multiprocessing/synchronize.py", line 17, in <module>
    import _multiprocessing
Run completed in 48710.69999999995ms
```